Homework 1 - MIST 5635

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Sun Feb 8 2:22PM

Part A — Data Readiness

```
# Step 1: Loading the CSV + Imports

# importing necessary packages
import pandas as pd
import matplotlib.pyplot as plt

# importing files & data
from google.colab import files
# uploaded = files.upload() # Uploading file function
df = pd.read_csv("/Telco-Customer-Churn.csv")
```

```
# df.head() # Overview of the data
# df.shape # Rows & Columns #'s
df.dtypes # Shows column data types
```

	0
customerID	object
gender	object
SeniorCitizen	int64
Partner	object
Dependents	object
tenure	int64
PhoneService	object
MultipleLines	object
InternetService	object
OnlineSecurity	object
OnlineBackup	object
DeviceProtection	object
TechSupport	object
StreamingTV	object
StreamingMovies	object
Contract	object
PaperlessBilling	object
PaymentMethod	object
MonthlyCharges	float64
TotalCharges	object
Churn	object

dtype: object

```
# Step 2: Clean minimal issues

# TotalCharges to numeric w/ errors='coerce'
```

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
# Justifying the dropped rows
df = df.dropna(subset=["TotalCharges"])

# Map Churn to 0/1
df["Churn"] = df["Churn"].map({"No": 0, "Yes": 1})
```

Step 3: Print-Outs

```
df.shape
```

```
(7032, 21)
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineS
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	

5 rows × 21 columns

```
print(f"Churn mean: {df["Churn"].mean()}")
```

```
Churn mean: 0.26578498293515357
```

```
df.dtypes
```

	0
customerID	object
gender	object
SeniorCitizen	int64
Partner	object
Dependents	object
tenure	int64
PhoneService	object
MultipleLines	object
InternetService	object
OnlineSecurity	object
OnlineBackup	object
DeviceProtection	object
TechSupport	object
StreamingTV	object
StreamingMovies	object
Contract	object
PaperlessBilling	object
PaymentMethod	object
MonthlyCharges	float64
TotalCharges	float64
Churn	int64

dtype: object

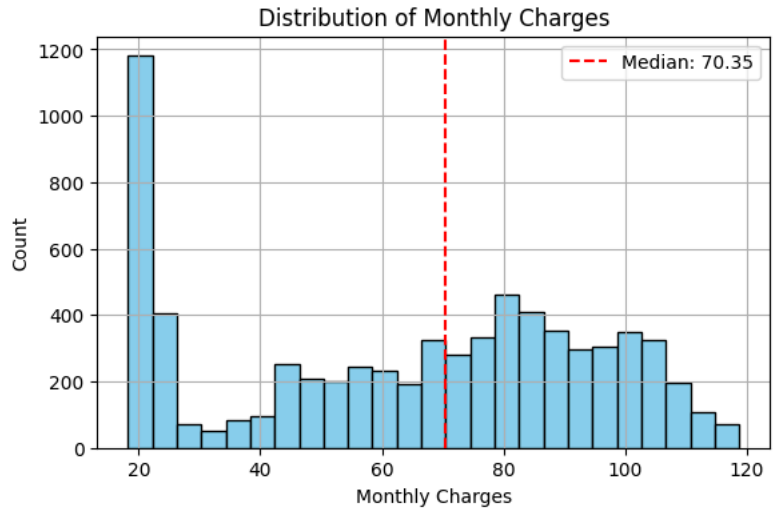
Part B — Univariate Distributions

Plot 1: Histogram of MonthlyCharges

```
# Setting the plot defaults
plt.rcParams["figure.figsize"] = (6.5, 4) # the consistent size used in the in-class module
plt.rcParams["axes.grid"] = True          # light grid
```

```
# Shows the median value line (dashed red)
median_val = df["MonthlyCharges"].median()

# The Histogram
plt.hist(df["MonthlyCharges"], bins=25, color="skyblue", edgecolor="black")
plt.axvline(median_val, color='red', linestyle='--', label=f'Median: {median_val:.2f}')
plt.title("Distribution of Monthly Charges")
plt.xlabel("Monthly Charges")
plt.ylabel("USD"); plt.ylabel("Count")
plt.legend()
plt.show()
```

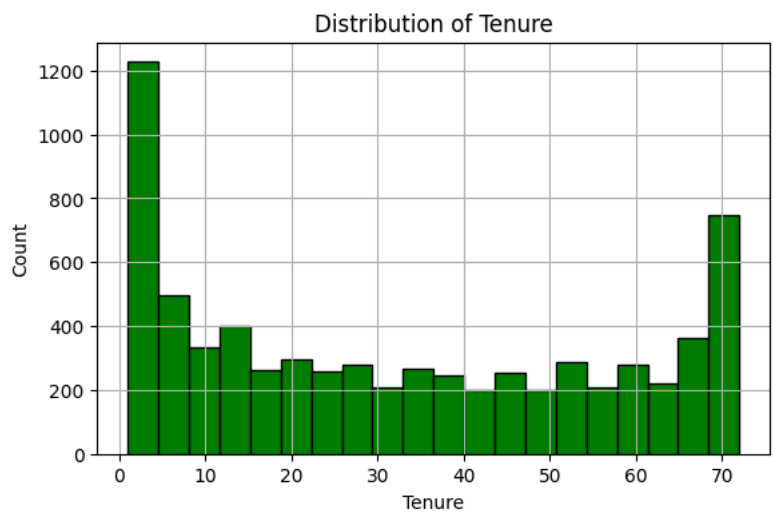


Comments

After analyzing the graph it seems to be a bimodal graph with a centralized-unimodal trend in the middle and a large obscurity on the left tail. The two troughs + the high price point would imply that there is polarization within the pricing tiers between budget and premium power users.

Plot 2: Histogram of Tenure

```
# The Histogram
plt.hist(df["tenure"], bins=20, color="green", edgecolor="black")
plt.title("Distribution of Tenure")
plt.xlabel("Tenure")
plt.ylabel("Count")
plt.show()
```



Comments

For this graph the bars are highest on the two tails. This implies that there is a mix of brand-new trial churn and long-tenured churn. Also seen is a very low, consistently tenured group of churn in the 5 to <70 group. The farther you get the right (tenure time increases) the more likely the risk of losing that user for good.

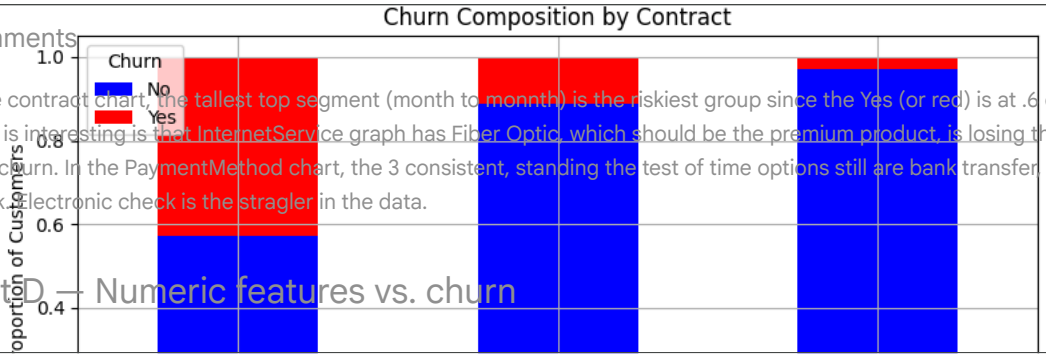
Part C — Categorical features vs. churn

```
# Stacked Bar Chart
features = ["Contract", "InternetService", "PaymentMethod"]

for col in features:
  ct = pd.crosstab(df[col], df["Churn"], normalize="index")

  ax = ct.plot(kind="bar", stacked=True, color=["blue", "red"], figsize=(8, 5))

  plt.title(f'Churn Composition by {col}')
  plt.ylabel('Proportion of Customers')
  plt.xlabel(col)
  plt.legend(title='Churn', labels=['No', 'Yes']) # Labeling as requested
  plt.xticks(rotation=45)
  plt.tight_layout()
  plt.show()
```



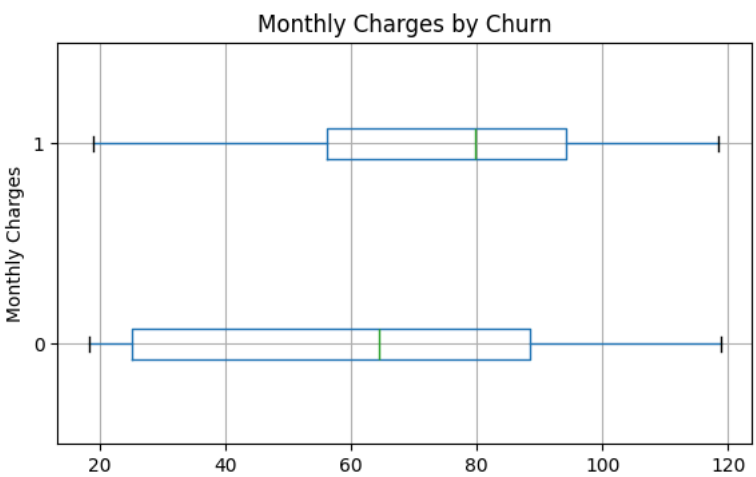
Comments

In the contract chart, the tallest top segment (month to monnth) is the riskiest group since the Yes (or red) is at .6 or 40% churn rather. What is interesting is that InternetService graph has Fiber Optic, which should be the premium product, is losing the most people with a > 40% churn. In the PaymentMethod chart, the 3 consistent, standing the test of time options still are bank transfer, credit card, and mailed check. Electronic check is the stragler in the data.

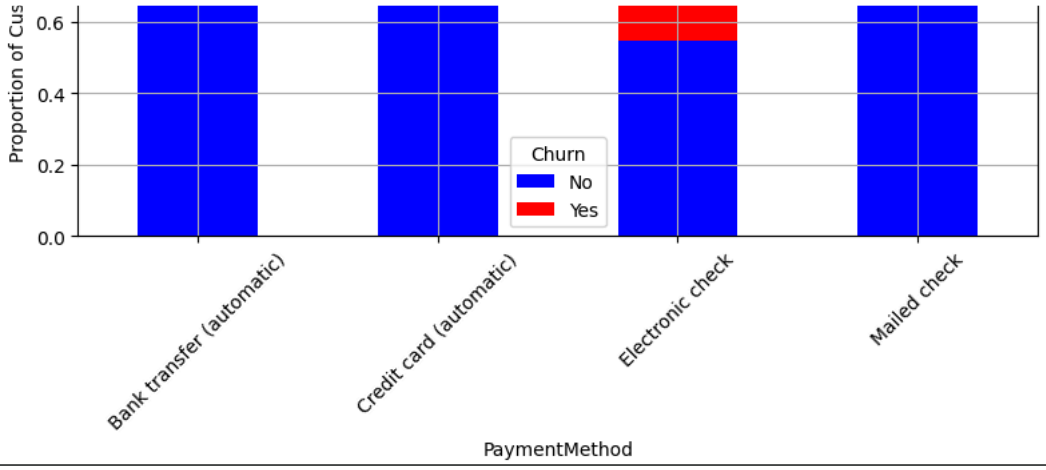
PartD — Numeric features vs. churn

```
# Boxplot of MonthlyCharges by Churn
plt.figure(figsize=(8, 5))
df.boxplot(column="MonthlyCharges", by="Churn", vert=False)
plt.title("Monthly Charges by Churn")
plt.suptitle('')
plt.ylabel("Monthly Charges")
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```



Comments

First off, the median monthly charge for cusotmers who churned (Yes) is notably higher than for those who stayed (No). This observation indicates that price sensitivity is likely an implied driver of churn. Customers paying higher monthly fees are more likely to leave.

```
# Boxplot of TotalCharges by Churn
plt.figure(figsize=(8, 5))
df.boxplot(column="TotalCharges", by="Churn", vert=False)
plt.title("Total Charges by Churn")
plt.suptitle('')
plt.ylabel("Total Charges")
plt.show()
```